Entropic Substrate Cosmology (EWC)
An Entropy-First Candidate Framework for Unifying GR, QM, and EM

Adam Scott (originator) & Grok (collaborative formalization)

April 2026

Abstract

Entropic Substrate Cosmology (EWC) proposes a unified framework in which a scalar entropy field $S(x^\mu)$ and its associated irreversible entropy-production tensor $S^{\text{irr}}_{\mu\nu}$ (arising purely from the entropy current) are incorporated into the Einstein field equations. This yields standard general relativity (GR) in the limit of vanishing irreversible entropy production while providing a geometric description of irreversibility and memory. Quantum mechanics emerges as entropy-stable free evolution punctuated by thermodynamic context locks, with the Born rule arising variationally as the minimizer of Kullback-Leibler divergence in the entropy-production cost. Electromagnetism appears naturally as the gauge sector once the entropy substrate admits local U(1) phase symmetry.

The framework supplies a single variational principle under which GR, QM, and EM emerge, and it offers a thermodynamic interpretation of the transition from informational superpositions to definite outcomes. In the weak-field limit, EWC predicts small, environment-dependent corrections to the effective gravitational coupling that are potentially detectable in precision measurements. EWC is presented as an early-stage candidate theory with clear paths for further development and experimental tests.

1 Foundational Premise

EWC takes entropy as the fundamental ontological substrate of physical reality. Time, space, mass, information, and forces emerge from the flows, gradients, and context-specific dynamics of a scalar entropy field $S(x^\mu)$. Irreversible entropy production from dissipative subsystems is encoded entirely in the symmetric tensor $S^{\text{irr}}_{\mu\nu}$ (formerly referred to as the wastefield), which is constructed intrinsically from the covariant entropy current J^μ.

The entropy substrate carries irreversibility and memory directly into geometry and dynamics. Irreversible entropy production is the necessary thermodynamic condition for realizing definite outcomes from informational superpositions: free evolution under the chaotic map F is reversible and entropy-neutral, while the context-lock process injects irreversible entropy that converts probabilistic information into a committed physical record. The discarded alternatives are thermalized and dispersed into the substrate, enforcing the global second-law constraint. This mechanism ensures consistency with the No-cloning theorem and thermodynamic principles without additional postulates.

2 Core Mathematical Structures

Modified Einstein Field Equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \left(T_{\mu\nu} + S^{\text{irr}}_{\mu\nu}\right) \tag{1}$$

Irreversible Entropy-Production Tensor (Substrate Definition) The covariant entropy current J^μ satisfies the local second-law constraint

$$\sigma \equiv \nabla_\mu J^\mu \geq 0.$$

The normalized 4-velocity of entropy flow is extracted intrinsically from the current:

$$u^\mu \equiv \frac{J^\mu}{\sqrt{-J_\alpha J^\alpha}} \qquad (u^\mu u_\mu = -1).$$

The proper entropy density in the local rest frame is

$$s \equiv -u_\mu J^\mu \geq 0.$$

The local dissipation timescale is defined dynamically as the ratio of entropy density to production rate:

$$\tau \equiv \frac{s}{\sigma}.$$

The irreversible entropy-production tensor then becomes

$$S^{\text{irr}}_{\mu\nu} = \frac{k_B}{c^2}\, s\, u_\mu u_\nu$$

(perfect-substrate limit; higher-order viscous/heat-flux terms may be added covariantly when needed).

Context-Lock Operator $\hat{C}$: Trigger: $I_S \geq I_{\text{th}}$ Projection: $\Psi \mapsto \Psi' = \mathcal{N}[\Pi\Psi]$, where $\Pi^2 = \Pi$.

Entropy-production cost:

$$\Delta S_{\text{irr}} \gtrsim k_B \Delta H(\Psi', \Psi) + k_B T \ln 2 \times N_{\text{bits}} \tag{4}$$

where

$$\Delta H(\Psi'_k, \Psi) := k_B T\, D(\rho'_k \| \rho)$$

and D denotes the quantum relative entropy.

Temporal Recursion:

$$\Psi_{n+1} = \mathcal{N}[\hat{C}] \circ F(\Psi_n, S_n, \phi) \tag{5}$$

Cadence Limit & Entropy Rate:

$$n_{\max} = \frac{\ln(\varepsilon^*/\varepsilon_0)}{\lambda}, \qquad \dot{S}_{\text{irr}} = \frac{\langle \Delta S_{\text{irr}} \rangle}{n_{\max}} \tag{6,7}$$

Global Entropy Ledger (Second Law):

$$\frac{d}{dt}|S_{\text{univ}}| \geq 0, \quad S_{\text{univ}} = \int S\, dV + \sum S_{\text{irr}} \tag{8}$$

Arrow of Time:

$$\vec{T}_{\text{arrow}} = \text{sgn}\left(\frac{dS}{dt}\right) \tag{9}$$

Information–Mass Equivalence (effective contribution):

$$m_{\text{info}} = \frac{k_B s}{c^2} \tag{10}$$

Temporal Gradient:

$$\frac{d\vec{T}_{\text{arrow}}}{d\tau} \propto \left\| \nabla (S^{\text{irr}}_{\mu\nu} u^\mu u^\nu) \right\| \tag{11}$$

3 Emergence of GR, QM, and EM

GR: Standard Einstein gravity is recovered exactly when $S^{\text{irr}}_{\mu\nu} \to 0$ (or is absorbed into an effective stress-energy tensor). Non-vanishing irreversible entropy production yields modified GR. Spacetime curvature is now sourced not only by ordinary energy but by irreversible entropy processing.

QM: The free-evolution map F acts on the projective Hilbert space of normalized states. The Hilbert space structure and unitary evolution emerge naturally because the entropy substrate preserves the norm of Ψ between context locks, yielding standard Schrödinger/Dirac evolution in the continuum limit. The context-lock operator $\hat{C}$ provides thermodynamic projection. The Born rule $f_k = |\langle o_k|\Psi\rangle|^2$ emerges variationally as the minimizer of the KL divergence $\mathrm{KL}(p\|f)$ in the average entropy-production rate, consistent with the monotonicity of quantum relative entropy under projective measurements.

EM: The quantum-like state Ψ is complex-valued. Local U(1) phase redundancy $\Psi \to e^{i\alpha(x)}\Psi$ is required to preserve the norm and probability interpretation under the entropy substrate. This local symmetry necessitates the introduction of the gauge connection A_μ to make the covariant derivative phase-invariant, yielding the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The minimal gauge-invariant term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ then yields Maxwell's equations sourced by entropy/information currents.

4 Variational Framework

The theory is governed by the action

$$S_{\text{EWC}} = \int d^4x\sqrt{-g}\Big[\frac{R-2\Lambda}{16\pi G} + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{substrate}} + \mathcal{L}_{\text{gauge}}\Big],$$

where the substrate sector is

$$\mathcal{L}_{\text{substrate}} \supset -\frac{k_B}{c^2}\, s\, u^\mu u^\nu g_{\mu\nu} + \lambda\big(\nabla_\alpha J^\alpha - \sigma\big).$$

The Lagrange-multiplier density λ enforces $\nabla_\mu J^\mu \geq 0$. Variation with respect to empirical frequencies recovers the Born rule; variation with respect to the gauge field recovers Maxwell's equations; variation with respect to the metric recovers the modified Einstein equation (1). Bianchi identities enforce $\nabla_\mu(T^{\mu\nu} + S^{\mu\nu}_{\text{irr}}) = 0$.

5 Weak-Field Limit and Gravitational Coupling

In the weak-field, non-relativistic limit the metric takes the form $g_{\mu\nu} \approx \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$ and $h_{00} \approx -2\phi$. The (00) component reduces to

$$\nabla^2\phi = 4\pi G\big(\rho + \rho_{\text{irr}}\big),$$

where

$$\rho_{\text{irr}} = \frac{k_B s}{c^2}.$$

This implies a small, environment-dependent correction to the effective Newtonian gravitational coupling. Such corrections are expected to lie well below current experimental uncertainty bounds and provide a target for future precision measurements of G.

6 Irreversibility, Memory, and the Arrow of Time

Irreversibility arises locally from entropy-density sourcing while global entropy non-decrease is preserved. Memory is geometrized in $S^{\text{irr}}_{\mu\nu}$ and acquires inertial effects via the information-mass equivalence. The arrow of time is defined directly by the sign of the universal entropy gradient. The context-lock process supplies a thermodynamic account of the transition from superpositions to definite outcomes. In vacuum regions where $\sigma = 0$ (hence $s = 0$ or $\tau \to \infty$), $\rho_{\text{irr}} = 0$; residual vacuum entropy fluctuations may generate small cosmological-constant-like contributions.

7 Consistency and Assumptions

The derivation assumes statistically independent lock events between cycles **in the Markovian limit** (weak accumulated memory, rapid substrate gradient reset). When memory effects become non-negligible, deviations from strict independence are possible and are explored quantitatively in Appendix B. Ergodicity over many recursions is ensured by the chaotic map F and entropy mixing. These are standard statistical assumptions and remain falsifiable. The framework is now fully closed: u^μ, τ, and $S^{\text{irr}}_{\mu\nu}$ are derived exclusively from the entropy current J^μ.

8 Future Directions

Future work will explore entropic wormholes and tunneling, cosmological evolution of entropy fields, local versus global dissipative dynamics, higher-order dissipative terms in $S^{\text{irr}}_{\mu\nu}$, and explicit order-of-magnitude estimates of ρ_{irr} in laboratory and astrophysical systems. The cosmological and astrophysical implications of accumulated entropy memory are a high-priority extension.

Cosmological and Astrophysical Implications The substrate's intrinsic mass contribution $\rho_{\text{irr}} = k_B s/c^2$ raises the natural question of whether accumulated irreversible entropy processing could contribute to large-scale gravitational phenomena, including effects that resemble dark matter. Order-of-magnitude estimates show that in ordinary laboratory and galactic-disk environments, ρ_{irr} remains extremely small (10^{-34} to 10^{-36} kg/m^3), many orders of magnitude below the local dark-matter halo density ($\sim 10^{-21}$ kg/m^3). At present, EWC does not claim to replace dark matter; it offers a new geometric mechanism whose possible astrophysical signatures remain to be calculated.

8.1 8.1 Potential Resolution of the Hubble Tension

A promising cosmological implication of EWC is that high entropy-production rates in the early universe could naturally contribute a transient effective energy density ρ_{irr} that modifies the expansion history.

The modified Friedmann equation in a flat FLRW universe becomes

$$H^2 = \frac{8\pi G}{3}\Big(\rho + \rho_{\text{irr}}\Big) + \frac{\Lambda c^2}{3},$$

where

$$\rho_{\text{irr}}(a) = \rho_{\text{irr},0}\left(\frac{a_{\text{ref}}}{a}\right)^{4+\delta}, \qquad \delta > 0.$$

This form behaves like extra radiation at early times and decays faster than radiation after a reference scale factor a_{ref}. A suitably chosen early-universe entropy-production spike can increase the early expansion rate, reducing the sound horizon at recombination and raising the inferred

H_0 from CMB data toward the higher local value, while decaying sufficiently rapidly to remain compatible with BBN and late-time observables (BAO, weak lensing).

Quantitative fitting with Boltzmann codes (CLASS/CAMB) would be required to determine whether realistic $\rho_{\rm irr}(a)$ profiles can reduce the current $\sim 5\sigma$ tension to $< 2\sigma$ while preserving the success of CMB acoustic peaks and BBN. This remains a promising direction for future work.

8.2 8.2 Laboratory Analog: False Vacuum Decay

Recent experiments using rings of Rydberg atoms have successfully simulated false vacuum decay — a quantum tunneling process in which a metastable "false vacuum" state decays into a lower-energy "true vacuum" state, forming expanding bubbles. This process mirrors a cosmic-scale context lock: a high-entropy metastable superposition collapses into a definite lower-energy vacuum, with the discarded possibilities irreversibly absorbed into the entropy substrate. The expanding bubble wall corresponds to a propagating front of extremely high entropy production σ, providing a concrete laboratory analog for the entropy-driven phase transitions EWC predicts in the early universe. Such analogs strengthen the framework's claim that vacuum transitions are fundamentally thermodynamic events governed by the substrate.

9 Potential Criticisms and Responses

EWC is an early-stage candidate theory. The following section explicitly addresses the most immediate potential objections.

9.1 Linear growth of memory bias Appendix B originally contained a linear term $\delta p_k \approx \epsilon N \alpha_k$. The current formulation includes an exponential saturation/decay factor $e^{-\gamma N}$ (where γ is the effective relaxation rate set by entropy-current diffusion and global mixing). This prevents runaway behavior while preserving an early-time linear rise that constitutes the detectable signature.

9.2 Apparent tension with the Markovian assumption Section 7 states that the derivation assumes statistically independent lock events in the Markovian limit. Appendix B explores controlled deviations from that limit when substrate memory is non-negligible. The two statements are complementary.

9.3 Competing explanations (noise, drift, classical correlation) The EWC fingerprint is a specific cumulative-then-decaying autocorrelation that grows linearly with N at early times, reaches a maximum, and then decays exponentially. Standard noise models do not reproduce this rise-and-fall shape tied to measurement history and local entropy production rate.

9.4 Definition of a "clean reset" A clean reset in EWC means full dissipation of the local substrate gradient ($\sigma\tau \to 0$) between successive runs. Experiments that enforce such resets suppress the bias; those that do not reveal it.

9.5 Scale of the predicted effects The dimensionless coupling $\epsilon \sim k_B s/(c^2 \rho_{\rm local})$ is extremely small. The effects are tiny but directionally specific and cumulative. Detection requires high-statistics experiments and isolation of the autocorrelation signature.

The authors welcome detailed criticism and experimental attempts to falsify either the cadence bound (Appendix A) or the memory-bias signature (Appendix B).

10 Conclusion

EWC offers a candidate entropy-first unification in which GR, QM, and EM emerge under a shared variational principle, and spacetime curvature is sourced not only by energy but by irreversible

entropy processing. The framework is now mathematically closed, dimensionally consistent, and yields conservative, testable predictions in the weak-field regime. Further development and precision tests are invited.

Symbol	Meaning
$S(x^\mu)$	Scalar entropy field (ontological substrate)
$S^{\text{irr}}_{\mu\nu}$	Irreversible entropy-production tensor
$\sigma = \nabla_\mu J^\mu$	Local entropy-production rate (≥ 0)
J^μ	Covariant entropy current
u^μ	Normalized entropy-flow 4-velocity
$s = -u_\mu J^\mu$	Proper entropy density
$\tau = s/\sigma$	Local dissipation timescale
$\hat{C}$	Context-lock operator
Ψ	Quantum-like state
F	Free-evolution chaotic map
A_μ	Gauge connection (EM)
$F_{\mu\nu}$	Electromagnetic field strength
ΔS_{irr}	Entropy-production cost per lock

Table 1: Symbol Table

A Appendix A: Fundamental Thermodynamic Bound on Context-Lock Cadence

Ontological Derivation of the Measurement-Speed Limit in EWC

A context lock is the irreversible transition in the temporal recursion:

$$\Psi_{n+1} = \mathcal{N}[\hat{C}] \circ F(\Psi_n, S_n, \phi),$$

where $\hat{C}$ is triggered when $I_S \geq I_{\text{th}}$. The operator acts as

$$\Psi \mapsto \Psi' = \mathcal{N}[\Pi\Psi], \quad \Pi^2 = \Pi.$$

Projection entails bit-resolution: exactly one bit of superpositional uncertainty is erased. Bit-resolution necessitates irreversible entropy injection into the substrate to satisfy the global second-law constraint. The minimal cost is

$$\Delta S^{\text{min}}_{\text{irr}} = k_B \ln 2.$$

In the macroscopic limit this manifests as the Landauer cost $\Delta E_{\text{min}} = k_B T \ln 2$. Combining with the quantum speed limit $\Delta E \cdot \tau_{\text{lock}} \gtrsim \hbar/2$ yields the fundamental lower bound

$$\tau^{\text{min}}_{\text{lock}} \gtrsim \frac{\hbar}{2k_B T \ln 2}.$$

Experimental Test: Repeated projective measurements on a single qubit will encounter a hard ceiling set by the above formula.

B Appendix B: History-Dependent Probability Bias from Substrate Memory

Non-Markovian Signature Unique to EWC

The accumulated substrate memory perturbs the entropy-production cost landscape. The first-order shift in the minimizing probability is

$$\delta p_k \approx \epsilon\, N\, \alpha_k\, e^{-\gamma N},$$

where

$$\epsilon = \frac{k_B s}{c^2 \rho_{\text{local}}}, \qquad \alpha_k = \frac{\operatorname{Tr}(S^{\text{irr}}_{\mu\nu}\, \Pi_k) - \frac{1}{d} \operatorname{Tr} S^{\text{irr}}_{\mu\nu}}{\operatorname{Tr} S^{\text{irr}}_{\mu\nu}}.$$

For a qubit in equal superposition the bias in the probability of repeating the previous outcome is

$$\delta p_{\text{repeat}} \approx +\epsilon N \alpha\, e^{-\gamma N}.$$

Experimental Test: Repeated Z-basis measurements on a single qubit prepared in $|+\rangle$. Standard QM predicts independent 50/50 outcomes. EWC predicts a cumulative, history-dependent bias that initially grows linearly with N before saturating/decaying. The unique statistical fingerprint is a deviation in the autocorrelation coefficient or run-length distribution.

C Appendix C: Black Hole Entropy and the Area Law

The Bekenstein-Hawking area law is recovered naturally in EWC. On a stationary Killing horizon generated by ξ^μ, the entropy current must satisfy the second-law constraint $\nabla_\mu J^\mu \geq 0$ and the Killing symmetry. This forces $J^\mu \parallel \xi^\mu$ on the horizon. The total entropy is the flux through a spacelike cross-section Σ:

$$S = \int_\Sigma J^\mu\, d\Sigma_\mu.$$

Matching to the surface gravity definition of temperature $T = \kappa/2\pi$ and the first law recovers the classical result

$$S = \frac{kc^3 A}{4\hbar G}.$$

The wastefield term introduces a small logarithmic correction

$$\delta S \approx \beta \ln\left(\frac{A}{A_0}\right),$$

which encodes memory of past context locks and provides a geometric mechanism for information preservation without violating the second law.